

Revised Generation Interconnection System Impact Study Report

For Queue Project Q18-0915

Sherbino I 138 kV
August 27, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0915. The Interconnection Customer proposes a 171.3 MW wind facility interconnecting at the Sherbino I 138 kV substation in the West Virginia area. The request was received on January 08, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 310 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$43,270,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: SHERBINO I - COAL 138.0 kV
Contingency: ANTELOPE VALLEY STA - HINTON AP SUBSTATION 138.0 kV L
Loading: 183.7% (Rating: 233 MVA)
DFAX: 82.0%
Pre-project loading: 123.5%
2. Facility: RHINELANDER PAPER - SNARR 138.0 kV line
Contingency: LOWMOOR - HINTON AP 345.0 kV
Loading: 160.7% (Rating: 203 MVA)
DFAX: 12.6%
Pre-project loading: 150.0%
3. Facility: SNARR - GLEN LYN 138.0 kV
Contingency: HINTON AP SUBSTATION - LOWMOOR 345.0 kV
Loading: 157.5% (Rating: 213 MVA)
DFAX: 12.6%
Pre-project loading: 147.4%
4. Facility: RADFORD ARMY AMMUNITION PLANT - ROCHESTER 138.0 kV L
Contingency: LOWMOR - HINTON AP 345.0 kV L
Loading: 157.0% (Rating: 183 MVA)
DFAX: 6.5%
Pre-project loading: 150.9%
5. Facility: MOUNT VIEW - PEYTON 138.0 kV
Contingency: HINTON AP - LOWMOOR 345.0 kV
Loading: 149.8% (Rating: 167 MVA)
DFAX: 7.5%
Pre-project loading: 142.1%

6. Facility: BUCKLEY - CELANESE ACETATE LLC 138.0 kV
Contingency: LOWMOOR - HINTON AP 345.0 kV
Loading: 149.5% (Rating: 198 MVA)
DFAX: 7.5%
Pre-project loading: 142.9%
7. Facility: ROCHESTER - SYSQUOC 138.0 kV
Contingency: LOWMOOR - HINTON AP 345.0 kV
Loading: 145.3% (Rating: 209 MVA)
DFAX: 6.5%
Pre-project loading: 140.0%
8. Facility: FT WHITE - RHINELANDER PAPER 138.0 kV L
Contingency: LOWMOOR - HINTON AP 345.0 kV L
Loading: 144.6% (Rating: 219 MVA)
DFAX: 7.5%
Pre-project loading: 138.7%
9. Facility: FT WHITE SUBSTATION - BUCKLEY SUBSTATION 138.0 kV
Contingency: LOWMOOR - HINTON AP SUBSTATION 345.0 kV line
Loading: 144.3% (Rating: 212 MVA)
DFAX: 7.5%
Pre-project loading: 138.2%
10. Facility: INGLES ST - SVP 138.0 kV
Contingency: LOWMOOR - HINTON AP 345.0 kV
Loading: 142.0% (Rating: 191 MVA)
DFAX: 6.5%
Pre-project loading: 136.1%
11. Facility: SVP - N CLAYTOR 138.0 kV line
Contingency: LOWMOOR - HINTON AP 345.0 kV L
Loading: 140.5% (Rating: 193 MVA)
DFAX: 6.5%
Pre-project loading: 134.7%
12. Facility: POMONA HTS - TALOGA 138.0 kV
Contingency: HINTON AP - LOWMOOR 345.0 kV
Loading: 140.2% (Rating: 252 MVA)
DFAX: 6.5%
Pre-project loading: 135.7%
13. Facility: NORTH CLAYTOR - RADFORD ARMY AMMUNITION PLANT 138.0 kV
Contingency: LOWMOOR - HINTON AP 345.0 kV
Loading: 137.7% (Rating: 211 MVA)
DFAX: 6.5%
Pre-project loading: 132.4%
14. Facility: HINTON AP 345/138kV TX
Contingency: MANATEE - HINTON AP 345.0 kV
Loading: 136.1% (Rating: 640 MVA)
DFAX: 21.2%
Pre-project loading: 130.4%
15. Facility: SOUTH VICKSBURG - CELANESE ACETATE LLC 138.0 kV
Contingency: LOWMOOR - HINTON AP 345.0 kV
Loading: 135.1% (Rating: 222 MVA)
DFAX: 7.5%
Pre-project loading: 129.3%
16. Facility: KIMBALLTON - COVADONGA 1014-1013-1012-1011 138.0 kV line
Contingency: LOWMOOR - HINTON AP 345.0 kV

- Loading: 134.6% (Rating: 218 MVA)
DFAX: 7.5%
Pre-project loading: 128.7%
17. Facility: S VICKSBURG - KIMBALLTON 138.0 kV Line
Contingency: LOWMOOR - HINTON AP 345.0 kV
Loading: 134.4% (Rating: 219 MVA)
DFAX: 7.5%
Pre-project loading: 128.5%
18. Facility: PEYTON - COVADONGA 1014-1013-1012-1011 138.0 kV
Contingency: HINTON AP - LOWMOOR 345.0 kV
Loading: 133.7% (Rating: 212 MVA)
DFAX: 7.5%
Pre-project loading: 127.7%
19. Facility: HARTMAN CREEK - MOUNT VIEW 138.0 kV
Contingency: HINTON AP - LOWMOOR 345.0 kV
Loading: 131.5% (Rating: 217 MVA)
DFAX: 7.5%
Pre-project loading: 125.5%
20. Facility: LAKE FORREST - SYSQUOC 138.0 kV
Contingency: LOWMOOR - HINTON AP 345.0 kV L
Loading: 130.9% (Rating: 243 MVA)
DFAX: 6.5%
Pre-project loading: 126.3%
21. Facility: LYTLE - LAKE FORREST 138.0 kV
Contingency: LOWMOOR SUBSTATION - HINTON AP 345.0 kV L
Loading: 128.7% (Rating: 256 MVA)
DFAX: 6.5%
Pre-project loading: 124.3%
22. Facility: MCCLAIN ENERGY FACILITY - COAL SUBSTATION 138.0 kV
Contingency: MANATEE - HINTON AP 345.0 kV
Loading: 127.7% (Rating: 367 MVA)
DFAX: 46.8%
Pre-project loading: 105.9%
23. Facility: LYTLE - POMONA HEIGHTS 138.0 kV
Contingency: LOWMOOR - HINTON AP 345.0 kV line
Loading: 126.6% (Rating: 270 MVA)
DFAX: 6.5%
Pre-project loading: 122.5%
24. Facility: COLTON - COAL 138.0 kV
Contingency: MCCLAIN ENERGY FACILITY - COAL 138.0 kV
Loading: 121.5% (Rating: 233 MVA)
DFAX: 73.3%
Pre-project loading: 67.6%
25. Facility: COLTON - CENTRAL JUNCOS 3202 138.0 kV
Contingency: MCCLAIN ENERGY FACILITY - COAL 138.0 kV
Loading: 120.6% (Rating: 233 MVA)
DFAX: 73.3%
Pre-project loading: 66.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SHERBINO I - COAL 138.0 kV	Rebuild line with higher-capacity conductor	\$3,171,000	60% (\$1,895,000)
RHINELANDER PAPER - SNARR 138.0 kV line	Rebuild line with higher-capacity conductor	\$4,521,000	28% (\$1,288,000)
GLEN LYN - SNARR 138.0 kV	Rebuild line with higher-capacity conductor	\$4,661,000	28% (\$1,327,000)
RADFORD ARMY AMMUNITION PLANT - ROCHESTER 138.0 kV L	Rebuild line with higher-capacity conductor	\$9,217,000	17% (\$1,582,000)
MOUNT VIEW - PEYTON 138.0 kV	Rebuild line with higher-capacity conductor	\$5,926,000	17% (\$999,000)
BUCKLEY - CELANESE ACETATE LLC 138.0 kV	Rebuild line with higher-capacity conductor	\$4,206,000	17% (\$709,000)
ROCHESTER - SYSQUOC 138.0 kV	Rebuild line with higher-capacity conductor	\$4,814,000	17% (\$826,000)
FT WHITE - RHINELANDER PAPER 138.0 kV L	Rebuild line with higher-capacity conductor	\$10,059,000	17% (\$1,696,000)
BUCKLEY - FORTWHITE 138.0 kV	Rebuild line with higher-capacity conductor	\$6,510,000	17% (\$1,097,000)
INGLES ST - SVP 138.0 kV L	Rebuild line with higher-capacity conductor	\$1,660,000	17% (\$285,000)
SVP - N CLAYTOR 138.0 kV line	Rebuild line with higher-capacity conductor	\$3,919,000	17% (\$673,000)
TALOGA - POMONA HTS 138.0 kV	Rebuild line with higher-capacity conductor	\$17,866,000	17% (\$3,067,000)
NORTH CLAYTOR - RADFORD ARMY AMMUNITION PLANT 138.0 kV	Rebuild line with higher-capacity conductor	\$2,246,000	17% (\$386,000)
HINTON AP 345/138kV Transformer Bank	Replace transformer with higher-rated unit	\$19,565,000	6% (\$1,117,000)
CELANESE ACETATE LLC - S VICKSBURG 138.0 kV	Rebuild line with higher-capacity conductor	\$4,783,000	17% (\$806,000)
KIMBALLON - COVADONGA 1014-1013-1012-1011 138.0 kV	Rebuild line with higher-capacity conductor	\$7,522,000	17% (\$1,268,000)
SOUTH VICKSBURG - KIMBALLTON 138.0 kV Line	Rebuild line with higher-capacity conductor	\$367,000	17% (\$62,000)
COVADONGA 1014-1013-1012-1011 - PEYTON SUBSTATION 138.0 kV	Rebuild line with higher-capacity conductor	\$8,632,000	17% (\$1,455,000)
MOUNT VIEW - HARTMAN CRK 138.0 kV Line	Rebuild line with higher-capacity conductor	\$2,900,000	11% (\$314,000)
LAKE FORREST - SYSQUOC 138.0 kV	Rebuild line with higher-capacity conductor	\$2,339,000	17% (\$401,000)
LYTLE - LAKE FORREST 138.0 kV	Rebuild line with higher-capacity conductor	\$3,585,000	17% (\$615,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MCCLAIN ENERGY FACILITY - COAL SUBSTATION 138.0 kV	Rebuild line with higher-capacity conductor	\$4,789,000	48% (\$2,314,000)
LYTLE - POMONA HTS 138.0 kV	Rebuild line with higher-capacity conductor	\$2,241,000	17% (\$385,000)
COLTON - COAL 138.0 kV	Rebuild line with higher-capacity conductor	\$13,052,000	100% (\$13,052,000)
COLTON - CENTRAL JUNCOS 3202 138.0 kV	Rebuild line with higher-capacity conductor	\$5,651,000	100% (\$5,651,000)
	Total allocated to project		\$43,270,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.